CockpitSentinel: Smoking Detection Report

Google Colab YOLO11s Training, Dataset Metrics & Edge Performance

Executive Summary: Trained YOLO11s on Google Colab (Tesla T4 GPU) achieving 96.89% mAP@50 and 93.58% recall for driver smoking detection. Exported to Intel OpenVINO FP16 for 22.7 FPS CPU inference.

1. Dataset Distribution & Annotations

The model was trained on the curated Roboflow dataset satyapal-gaikwad/driver-smoking-phoneuse-earphone-detection-9nlii-oscoe comprising 1,253 labeled driver frames and 1,552 bounding boxes.

Class ID	Class Label	Train Boxes	Val Boxes	Total Annotations	Target Role
0	earphone	333	122	455	Secondary distraction
1	person	191	0	191	Driver cabin context
2	smoke	613	70	683	**Primary target (Cigarettes/Vapes)**
3	using_phone	193	30	223	Secondary distraction
All	Total	1,330	222	1,552	Complete Dataset Volume

2. Training Setup & Hyperparameters (Google Colab)

- **Hardware:** NVIDIA Tesla T4 GPU (16 GB VRAM) on Google Colab
- **Architecture:** Pretrained YOLO11s (100 fused layers, 9.4M parameters, 21.4 GFLOPs)
- **Optimizer & Epochs:** AdamW, lr0=0.001, 60 epochs (duration: 1,068.6 s = ~17.8 min)
- **Resolution:** 640x640 with RandAugment, Random Erasing (0.4), and Mosaic (closed at epoch 50)

3. Quantitative Accuracy & Validation Results

Evaluated on the independent 201 validation images (222 ground-truth instances):

Target Class	Instances	Precision	Recall	F1-Score	mAP@50	mAP@50-95
smoke (Smoking)	70	92.91%	93.58%	93.24%	96.89%	59.14%
using_phone	30	90.71%	86.67%	88.64%	90.41%	42.34%
earphone	122	90.43%	77.49%	83.47%	85.06%	35.82%
Overall (All)	222	91.35%	85.91%	88.55%	90.78%	45.77%

4. Edge CPU Latency & Deployment Benchmarks

- **PyTorch CPU:** 343.4 ms/frame (2.9 FPS)
- **Intel OpenVINO FP16:** 44.0 ms/frame (22.7 FPS) → **7.81x faster**
- **Heuristic Box Filter:** Discards smoking boxes > 25% frame area to eliminate ambient glare.
- **Temporal Confirmation:** Requires 2.0s sustained detection before firing audible/visual alerts.